

02LQD Specification and Simulation of Digital Systems

2019/02/11

1. Combinational circuit design (3 points)

In a base-3 numbering system, given 2 inputs A and B, a half-adder (no carry-in) computing the Carry (out) and the Sum is defined by the following table:

<i>A</i>	<i>B</i>	<i>Carry</i>	<i>Sum</i>
0	0	0	0
0	1	0	1
0	2	0	2
1	0	0	1
1	1	0	2
1	2	1	0
2	0	0	2
2	1	1	0
2	2	1	1

Encoding A, B and Sum on 2 bits a_1a_0 , b_1b_0 , s_1s_0 :

- draw the truth table for the half-adder, defining proper output size
- design in VHDL the combinational circuit implementing the truth table according to the following description styles:
 3. dataflow using conditional signal assignments (WHEN ELSE statements)
 4. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM design (12 points)

A **median filter** removes **lone 0s** in the input stream by changing each **lone 0** to a 1 on the output. A lone 0 is a 0 « sandwiched » between two 1s.

Example (lone 0s in boldface)

input stream: 0100**10**10111**10**1001

output stream: 010011111111001

Note that the output stream is the (modified) input stream 2 two clock ticks later.

Design a Moore FSM with one serial input X and one serial output Z compliant with the specifications. Reset is asynchronous.

- Draw the state diagram (7 points)
- Design the FSM resorting to the VHDL behavioural description style with 2 processes (5 points).

3. FSM-D design (15 points)

Multiplication is a complex operation. A non-optimized algorithm is based on repetitive addition. For example $7 \times 5 = 7 + 7 + 7 + 7 + 7$. Design a FSM-D to perform multiplication of pure binary numbers by repeated addition. Inputs a and b are on 8 bits. Define a proper size for the output z. Computation starts when a start signal is asserted for one clock cycle. When computation is over the ready signal is asserted for one clock cycle. Reset is asynchronous. Values for output z are available at each computation step, but they make sense only when ready is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**5 points**)
2. Design a FSM-D (**10 points**):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- no later than Thursday February 14 1.59 pm**
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
 - once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.